jasa-acs-qc

Quality-control audit for JASA ACS and other statistics submissions

What it does. Given a manuscript PDF and its supplement, Claude runs a set of scripted, mechanical checks and writes a Word report of **verifiable defects**: things that are wrong, missing or inconsistent as printed, which the authors cannot dispute once shown. Every finding is checked before it is written, and the report also lists what was checked and found correct.

What it does not do. It does not judge significance, novelty or journal fit. Use it alongside an editorial review (for example the companion `jasa-acs-review` skill, which covers grammar and the five ACS criteria).

Module	Question it answers	Typical yield
A Main ↔ supplement	Do the two documents agree with each other?	Dangling references, theorem/proof assumption mismatches, caption–table mismatches
B Numbers & claims	Do the printed numbers agree with themselves when recomputed?	Transposed digits, wrong “true” values, p-value orderings that cannot hold
C ACS structure	Does the paper have the shape ACS expects? (counted, not judged)	Unanswered scientific questions, data-analysis share below 25%
D Proof check	Are the theorems true as stated?	Missing hypotheses, sign errors, objects that do not exist for the estimator used

Guiding principle
Reading finds the loud errors; scripted, mechanical passes find the ones that decide whether a theorem or a table is true.

1. Getting started

Install

- You receive the skill as `jasa-acs-qc.skill` (a zip containing one `SKILL.md`).
- Add it to Claude as a custom skill by uploading that file in Claude's skill settings. Once added it appears as `jasa-acs-qc` and Claude uses it automatically when a request matches.
- Nothing else to install by hand: Claude installs what it needs (`poppler-utils` for pdftotext/pdftoppm, `numpy`, `python-docx`) in its workspace.

What to provide

Input	Required?	Notes
Main manuscript PDF	Yes	Journal proof PDFs are fine; the "For Peer Review Only" watermark is handled.
Supplement PDF	Strongly recommended	Module A needs it. Without it, Modules B–D still run on the main text.
Manuscript ID	Optional	Used in the report title and file name (e.g. JASA-ACS-2026-0123).
Scope	Optional	Say "proofs only", "numbers only", "skip Module C", or "split the report" if you want.

How to ask

Attach the PDFs (or point Claude to the folder that holds them) and use plain language. Any of these trigger the skill:

```
QC this submission: main.pdf and supp.pdf. ID JASA-ACS-2026-0123.
Run the consistency audit between the paper and its supplement.
Check the proofs in Appendix B against the theorems in Section 3.
Verify the numbers in Tables 2 and S4 — something looks off.
Audit this manuscript; give me separate consistency and proof-check reports.
```

What you get back

- `<ID>_QC_report.docx`, saved next to the manuscript (or two files, `<ID>_consistency_audit.docx` and `<ID>_proof_check.docx`, if you ask).
- A short chat summary: severity counts and the two or three findings that matter most.

Typical run

1 Extract	2 Inventory	3 Modules A–D	4 Verify	5 Report
Text from both PDFs (raw + layout); page offset	Theorems, assumptions, tables, figures, references	Scripted passes; code for every numeric check	Render pages, grep absences, recompute	Word report + chat summary

2. Ground rules the skill always follows

These rules are what make the findings hold up. If you edit the skill, keep them.

Rule	Why
Render the page before asserting any equation-level finding	Text extraction drops brackets and misplaces subscripts. An equation that looks wrong in extracted text is often correct on the page. Applies to typo comments too.
Grep before writing “the paper never does X”	A wrong absence claim discredits the whole report. Both files are searched for the term and its variants.
Look for the authors’ own competing usage first	The strongest finding is “equation (A) and equation (B) cannot both hold”, citing two places in the paper.
Check whether two numbers come from different experiments	Different replicate counts across tables are usually correct, just under-captioned. Downgrade rather than defend.
Simulate when a derivation or algorithm step is disputed	Implementing the paper’s own iteration on a tiny synthetic instance produces a number nobody can argue with.
Keep the manuscript confidential	Work only on local copies; never upload the text elsewhere or search the web for the manuscript.

Supplement page numbers
Supplements often print page numbers continuing from the main text (printed p.48 may be PDF p.13). The skill computes this offset once and says which numbering the report uses.

3. The four modules

Module A — Main ↔ supplement consistency

Pass	What is checked	Finds
A1 Pointers	Every cross-reference in both directions resolves to an object that exists and says what the citing text claims; every supplementary object is cited at least once.	Dangling references, wrong targets, orphaned tables/figures
A2 Theorem vs proof	Assumptions in each theorem statement vs those the proof actually uses, term by term. *Highest-yield pass.*	Conditions added or dropped in the proof; lemmas proved under stronger conditions than claimed
A3 One-to-one displays	Per-row figures vs table rows: count, order, identifiers; caption numbers vs cells; shared columns digit-for-digit.	Mismatched or duplicated figures; sign errors in captions
A4 Reference lists	Main vs supplement bibliographies and in-text citations.	Works discussed but never listed; duplicate author–year entries
A5 Shared descriptions	Cohort size, covariates, prevalence, tuning constants, replicate counts, notation.	Side-by-side discrepancy tables

Module B — Numerical and claims verification

Check	How	Example of what it catches
B1 Marginal reconciliation	Two tables partitioning the same cohort must have matching weighted marginals and sum to N.	Descriptive tables that cannot both be right
B2 SE vs rate	Infer replications $R = p(1-p)/SE^2$ from the bulk; flag entries that do not fit.	Coverage printed 0.552 (0.010) where the SE fits 0.952
B3 Convergence rate	log(length ratio) / log(n ratio) against the rate the paper's own theory implies.	Main table shrinking like n^{-1}, supplement like $n^{-1/2}$
B4 Recompute the statistic	Rebuild the paper's statistic from reported estimates; count discordant pairs with the printed p-value order; strict-dominance check.	Results table inconsistent with the printed parameters
B5 Stated "true" values	Derive the true parameter from the stated generator; compare with the estimation table at largest n.	Confusing B(a,b), 1/B(a,b) and 1/(αB(a,b))
B6 Text arithmetic	Percentages vs counts, flow-diagram totals, CIs vs estimates, exp(β) vs reported ORs.	Routine but frequent slips

Module C — ACS structure (counted, not judged)

- **Questions → answers.** Each scientific question in Section 2 is traced to the table, figure or subsection that answers it.
- **Data-analysis share.** Pages of real-data analysis as a percentage of the main text (ACS expects more than 25%).
- **Simulation placement.** Pages of simulation in the main text (ACS expects at most about one page of summary).
- **Model checking, sensitivity, baselines, uncertainty.** Presence or absence, grep-verified.
- **Reproducibility.** Code and data availability statement; seeds and software versions where results depend on them.

The verdict on these criteria stays with the reviewer; the skill only supplies the counts.

Module D — Proof check (when there is a theory appendix)

#	Check	#	Check
1	Hypothesis reconciliation — every condition used vs every condition stated; flag undefined "≫"	8	Specification vs approximation — exact model membership silently assumed
2	Substituted bounds — re-derive each bound from the cited condition	9	Concavity/uniqueness — profile out the nuisance before claiming a unique maximiser
3	Units/dimensions — e.g. pseudo-inverse gives $N(0, V^{\dagger})$, not $N(0, I)$	10	Tangent-space support — efficiency needs the declared region to equal the score's support
4	Algebraic identities — substitute and expand; sign errors, spurious factors	11	Upper bound vs improvement — comparing two upper bounds proves nothing
5	Limits/quantiles — evaluate literally; boundary cases defined	12	Monotone descent — only exact block or MM steps are guaranteed
6	Decorations — hats, stars, indices, matrix sizes	13	Imports — paraphrased external theorems vs their exact numbered statements
7	Existence — does the object (minimiser, transport map) exist for the estimator actually used?		

4. Reading the report

Severity scales

Modules A–C	Module D (proofs)
Critical — changes a reported result or the truth of a stated theorem	**Error** — false as written
Major — must be fixed for the paper to be correct or reproducible	**Gap** — hypothesis missing, or a step asserted without argument
Minor — presentation, captioning, cross-reference	**Defect** — typographic
Verified consistent — checks that passed (mandatory)	**Verified as correct** — identities, substitutions, rates reproduced

Report layout

Section	Contents
Title block	Manuscript ID, title, files audited with page counts, date
Count box	Findings by severity, plus the number of verified-consistent checks
A, B, C, D…	One lettered table per module run: # · Severity · Location · Finding · Suggested fix. Side-by-side comparison tables sit under the finding they support.
Verified as correct / consistent	Roughly a third of the report. Specific ("all 27 captions match Table S7"), so authors know what not to touch and the audit is visibly systematic.

Section	Contents
Priority for the revision	The 3–6 items to fix first, drawn from this paper's findings only
Methods note	One paragraph: passes run and tools used

Why the "verified" section matters
It tells authors where not to spend revision effort, shows the audit was systematic rather than opportunistic, and makes the critical findings much harder to wave away.

5. Tips and troubleshooting

Situation	What to do
No supplement	Ask anyway; Module A is skipped and the report says so. Modules B–D run on the main text.
Scanned (image-only) PDF	Text extraction returns little. Ask Claude to OCR first, or supply the original PDF/LaTeX.
Odd stray letters in quoted text	Usually the "For Peer Review Only" watermark interleaved by extraction, not author typos.
A finding looks wrong to you	Ask Claude to show the rendered page or the code that produced it. Every finding should be reproducible on request.
Long theory appendix	Ask for "proof check only, Theorems 1–3 first" and continue in a second request.
You want two separate files	Say "split the report"; you get a consistency audit and a proof check.
You also want an editorial verdict	Run an editorial-review skill separately; this skill deliberately stays factual.

The skill's final checklist (run before delivery)

- ☐ Every equation-level finding checked on a **rendered page**
- ☐ Every absence claim **grepped** in both files
- ☐ Every numerical finding **recomputed by code**
- ☐ Table differences checked for a "different experiment" explanation
- ☐ Page numbering stated (printed vs PDF)
- ☐ Count-box totals equal the number of table rows
- ☐ Priority list drawn from this manuscript only
- ☐ The .docx opened and confirmed to render

Confidentiality reminder
Manuscripts under review are confidential. Share this skill freely, but never share manuscripts or reports produced with it outside the review process.